Contrast.

The concept of contrast

The concept of contrast in visual art is quite universal, and therefore it is filled with specific content depending on the context.

There are three types of contrast in physiological optics:

- according to the brightness/Brightness
- by saturation
- by color tone

Tonal contrast/Luminance contrast

Light contrast is of fundamental importance in the visual arts.

The light-density contrast is a total value that consists of differences in the reflection coefficients of various surfaces and the brightness of their illumination, as well as the total photochemical contrast of the end-to-end photographic process.

K - contrast, the ratio of two brightness levels.

B_1, B_2 - brightness.

When $B_1 > B_2$,

$$K = \frac{B_1 - B_2}{B_2}$$

large light contrast: $K > 0.5$
medium light contrast: $0.5 > K > 0.2$
low light contrast: $0.2 > K$.

A table of brightness intervals for some shooting objects. [1]

Table of reflectivity of various surfaces. [2]

Optimal Visual Contrast (OVC) [1]

We have different brightness in different parts of the scene. The contrast of a scene is determined by the ratio of the brightness of the darkest and lightest areas. The luminosities of the image are located within this ratio.

Optimal visual contrast is a value in the range of such ratios from 1:40 to 1:60 (+- 2-3 stops from mid tone [3]) - in this contrast range, the human eye is able to simultaneously distinguish details, for example textures, both in the darkest

areas and in the lightest. When the contrast level of a scene is increased, for example, to 1:200, we are no longer able to simultaneously perceive details in all areas of the image. That is, OVC is an anthropometric module, the contrast of any work of fine art must be correlated with its magnitude, any tonal solution must take this value into account as a module embedded in our visual perception.

You need to feel the optimal visual contrast well: when choosing the composition of the frame, determining the tonal figure-background ratios and organizing lighting in the frame, you need to feel that the contrast of the frame correlates with the OVC in a certain way, since it cannot be neglected. We must either maintain optimal visual contrast, or consciously violate it, knowing exactly what we are sacrificing - details in the shadows or details in the lights - because uncertainty in solving this issue leads to the inexpressiveness of the image.

There is a technique (when creating an image [4]) that creates separate lighting schemes (separate light schemes) of different parts of the image, thus allowing you to provide information that would be obtained when looking at an object with a restructuring (displacement of the point of visual adaptation) of the light sensitivity of the eye (occurs in the brightest part[5]).

Engravings are an interesting and illustrative example for considering and understanding OVC. The craftsmen who create them are able, working in the contrast range of 1:40, to create objects that in nature significantly exceed this contrast ratio. In a "special way" they compress the breadth of the luminance range of nature to the value of the OVC, redistribute the light-density ratios. One engraving contains several levels of light-density adaptation, the optimal visual contrast is applied several times differently to different parts of the object, while the OVC remains constant, and the level of adaptation changes all the time, which we cannot do using black-and-white film with exactly the same latitude of 1:40. Durer’s engraving[2] contains a complete perceptual picture of nature, as if we were looking at different places in a time interval and our visual apparatus would adapt to their brightness accordingly. Otherwise, [3] The OVC is intentionally violated by the master, who uses only black and white in the execution of the engraving, this is perceived by our perception as an image of an object whose contrast exceeds the OVC, this is a strong means of expression.

"No color can be lighter than white and darker than black, that is, color perception and color reproduction are, as it were, inside the rigid framework of tone reproduction, the boundaries of which correspond to OVC. Whether we like it or not, we have to admit that the dictatorship of OVC in the visual arts is inevitable, as it is determined by the physiological peculiarity of our vision. <...> In the visual arts, the main thing is to

correctly convey the relationships: scale, tonal, color or any other. But before you can begin to convey these relationships, you need to see and understand them."

"Color and contrast. Technology and creative choice" by V.Zheleznyakov

From the consideration of the engravings, we can divide the frames into two types: those brought to OVC and those not brought. As a rule, the frames shown for OVC contain more details, as a result, they are more informative. This applies to both black and white and color frames. When editing an episode, we must monitor the consistency of the contrast of the adjacent frames, the correspondence, or inconsistency, of its OVC. This is especially important when editing black-and-white and color frames within a single episode, if the transition from an achromatic to a color frame, or vice versa, is intended to be invisible to the viewer. Thus, the tonal unity of an episode, or even the entire film, depends on maintaining the level of white and the level of black in all frames. This applies to both black and white and color frames. However, it should be remembered that violation of this principle, as well as violation of OVC, is also a very strong expressive means.

Disproportionate perception

There is a concept of non-linear perception of a linear (equally contrasting) light-density range.

Dependence of sensory arousal (B) on perceived lightness (b - brightness):

$$B = log(b)$$

when increasing b to a certain high value, we get:

$$B = const$$

Logarithmic relationship between arousal and sensation when perceiving an equidistant series of increasing brightness (in accordance with the Weber-Fechner law)[4]

The logarithmic relationship between arousal and sensation when perceiving an equidistant gray scale (an analog of a real object with OVC)[5]

The most important thing in exposure calculations is to correctly correlate the visual perception of contrast with the perception (and transmission) of this contrast through a photographic/cinematographic process or other reproducing system.

A change in the sensation gradient when perceiving an equal-contrast object with a brightness range significantly greater than the OVC value.[6]

Due to the nonlinear perception of the equal-contrast light series, painters such as Frans Hals used different shades of black to work out details in deep shadows in order to increase the threshold distinctness for our eyes. This substitution of brightness contrast with color is well perceived by our vision, although there will be no difference between them when measured by an instrument (brightness meter). This is a widespread technique in the visual arts.

"With the help of chiaroscuro, Rembrandt not only highlighted the most important places of the painting, but also determined the temporal sequence of its perception. When you approach, for example, the painting "The Return of the Prodigal Son," you first pay attention to the brightly lit part of the painting, which for a moment seems to dazzle and deprive you of the ability to see anything else. Then, in the semi-darkness, you gradually begin to distinguish the surrounding figures, and finally, after adapting your eyes, you penetrate into the very depths of the shadows and distinguish more and more new details there. <...> Rembrandt perfectly felt the value of optimal visual contrast as an anthropometric module, as evidenced by his engravings, in addition, he excellently used the property of nonlinearity in the perception of contrast by the eye."

"Color and contrast. Technology and creative choice" by V.Zheleznyakov

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Color contrast (by saturation/color tone)

With regard to color contrast, the following types of contrast are possible, which are the consequences of color induction:[6]

1. Simultaneous contrast [7]

Simultaneous color contrast is well observed when two color shades are perceived simultaneously with neutral gray. In this case, the gray field located on one of the color fields acquires the shade of the other color field.

2. Sequential contrast [8]

Sequential color contrast is similar to simultaneous contrast, but only in the form of a sequential image. It is observed if you look at a colored field for 20-30 seconds, and then look at a colorless (white) field, then this field will turn into an additional (complementary) color located at the opposite end of the color wheel.

3. Complementary contrast [9]

The complementary contrast is shown by the proximity of additional colors. The feeling of saturation of each of them increases, however, only as long as they occupy a relatively large area in the field of view. When viewed from a great

distance, the law of additive mixing comes into effect, and the component complementary colors are perceived as a single gray spot. Pointillist artists who created images in their paintings from colored dots and small strokes, saturated in color, were well aware of this feature of perception and took it into account in their work.

4. Tint contrast [10]

The tint contrast is clearly evident when all three complementary colors are located side by side. At the same time, their shade perception becomes difficult.

5. Qualitative contrast [11]

A qualitative contrast is observed when comparing different brightness and saturation levels of the same color tone. It is widely used in the grisaille technique[7], when a watercolor drawing is executed with one paint of any one color. As an expressive means, qualitative contrast contrast is used when shooting portraits and nudes, when all the richness of human skin tones is reproduced on the face and figure of the model.

6. Quantitative contrast [12]

A quantitative contrast occurs between two objects (or objects) painted in different colors if their masses in the frame are very different from each other, and there must be some kind of psychological balance between these spots, which are different in color and size. Usually this type of contrast is combined with a kind of complementary contrast, when a small spot enhances the sound of the color field of the entire image. This is the so-called color accent in the frame. This technique has been used in painting for a very long time and very successfully. In the landscapes of the old masters, you can often see a small figure in a red suit or even just a red cap against a background of greenery - this small red spot immediately enlivens the greenery in the whole picture.

7. Saturation contrast [13]

The saturation contrast is especially noticeable when comparing achromatic colors with chromatic ones, while the lightness of the background on which the color is placed is of great importance: on a dark background, the color looks brighter and more saturated than on a light one. This type of contrast is very widely used in practice, enabling the artist to achieve an intense sound of color.

Optical attachments that change the contrast of the object being photographed

Three main types exist:

- change the color of the image
- change the optical pattern (sharpness)
- change the contrast

The second and third types of optical filters reduce the light contrast of the image. There is a quadratic dependence of the total sharpness of the lens-camera system (resolution) on the contrast of the image. This means that if the contrast is reduced, for example, by half, the sharpness will decrease four times. This is especially noticeable on small details in the frame and textures. To smooth out this effect, lighting is required, the contrast in nature at which exceeds OVC. One of the ways to achieve this effect is a "precision"[8] lighting system using lighting devices with Fresnel lenses[9].

When using optical filters during shooting, you should always arrange the frame and lighting after installing the filter on the camera, that is, the composition should be selected taking into account the optical nozzle.

The third type of nozzles is specialized to change the contrast of the image without affecting the optical pattern. Most often, these filters highlight deep shadows, which allows you to bring the contrast of the frame to the OVC value, while almost without changing the white level (LC - Low Contrast). There are also filters of the same type ("Soft Contrast", or "black filters"), which exclude the formation of any veil in the shadows, contributing to the production of a so-called diamond negative, when the discernibility of small details in the shadows is maximum. The mechanism of their action lies in the fact that, without affecting the density of the negative in deep shadows, they reduce the density in light and especially in glare, which results in a decrease in image contrast and a special brilliance, which increases the visibility of small details. These light filters have the side effect of acting as neutral gray due to the fact that they transmit less light than pure optical glass. In essence, their effect is similar to that of black grids, but without the "Kreuz effect", i.e. they do not form rays from light sources. The third type of optical filters is very sensitive (force of action) to extraneous light, therefore, lens protection with a lens hood is necessary to work with them.

Any optical attachments reduce the contrast of the image to one degree or another, this is not always a bad thing, but it is necessary to take this aspect into account when shooting. We also often do not know to what extent the contrast is reduced by the labeling filter, so it is worthwhile to conduct a series of tests in advance and find out this value for the optical nozzle that we want to use.

Literature:

- "Zheleznyakov V.N. Color and contrast. Technology and creative choice. Moscow: VGIK, 2001"
- "Arnheim R. "Art and visual perception"
- "Itten I. The Art of Color"

Notes and tables:

[1] Table of brightness intervals for some subjects:

Subject being shot	**Subject brightness range**
The Earth’s surface visible from an airplane in winter	1:6 - 1:10
Landscape without foreground in fog	1:2 - 1:3
Landscape without foreground on a cloudy day	1:5 - 1:10
Landscape without foreground in direct sunlight	1:10 - 1:30
Landscape without foreground against the light	1:20 - 1:40
Landscape with foreground in direct sunlight	1:20 - 1:60
Landscape in sunny weather, if the foreground is in shadow	1:100–1:300
Landscape with the sun in the frame	1:200000
Urban landscape without foreground in cloudy weather	1:5 - 1:10
Urban landscape without foreground in sunlight	1:10 - 1:40
Narrow darkened streets with individual buildings illuminated by the sun	1:100–1:500
Dark buildings against the light against the sky	1:100 - 1:200
Dark spans and arches of bridges and gates with brightly lit foreground	1:1000–1:10000
Groups of people on a sunny day (depending on the color of clothing and the direction of light)	1:20 - 1:300
Groups of people on a cloudy day	1:10 - 1:60
Portrait of a man with blond hair on a background illuminated by sunlight	1:10 - 1:12
Portrait of a man with dark hair on the background of an open landscape in sunlight	1:20 - 1:100
Interiors shot on film equipment (depending on the nature of the windows and lighting)	1:20 - 1:100
Interior view of the room (without windows in the frame)	1:8 - 1:12
Interior view of a bright room, shot against lighted windows	1:100–1:500

Subject being shot	Subject brightness range
Interior view of a dark room, filmed against brightly lit windows without illumination	up to 1:100000

[2]

Surface/Material	Reflectivity (albedo)
Fresh snow	0.80-0.95
Old/wet snow	0.40-0.70
Clean ice	0.30-0.50
Water (low Sun/ripples)	0.05-0.20
Water (high Sun, smooth surface)	0.02-0.06
Light sand	0.30-0.45
Dark sand	0.15-0.30
The ground is dry and light	0.25-0.40
Dark moist soil	0.08-0.20
Grass/meadows	0.20-0.30
Coniferous forest	0.08-0.15
Deciduous foliage	0.15-0.25
Fresh dark asphalt	0.05-0.10
Light concrete	0.25-0.35
Brick	0.20-0.30
Roofs are dark (tiles/bitumen)	0.05-0.15
The roofs are light (metal/coverage)	0.50-0.75
Limestone/light marble	0.40-0.60
Gypsum/white plaster	0.60-0.85
White paper	0.70-0.90
Matte black fabric	0.02-0.05
The fabric is white	0.60-0.85
Polished aluminum (specular)	0.80-0.90
Brushed aluminum	0.50-0.70
Polished steel	0.55-0.65
Oxidized steel/brushed	0.30-0.50
Polished copper	0.70-0.85
Polished gold	0.70-0.85
Polished silver	0.85-0.95

[3] In broadcast/television interpretations, an operating interval equivalent to ≈1:32 for the gray scale is often used, which explains the more "narrow" practical norm in signal mixing and monitoring.

[4] In cinema and photography, this is achieved by using a large number of lighting devices, each of which highlights its own area. Valeurs (tonal values) are used in painting.

[5] From practical observations, it is clear that our eye does not take any brightest spot for the white level, but only one that occupies a certain area on the retina.

[6] The phenomenon of color induction is a change in the characteristics of a color under the influence of observing another color or, more simply, the mutual influence of colors on each other.

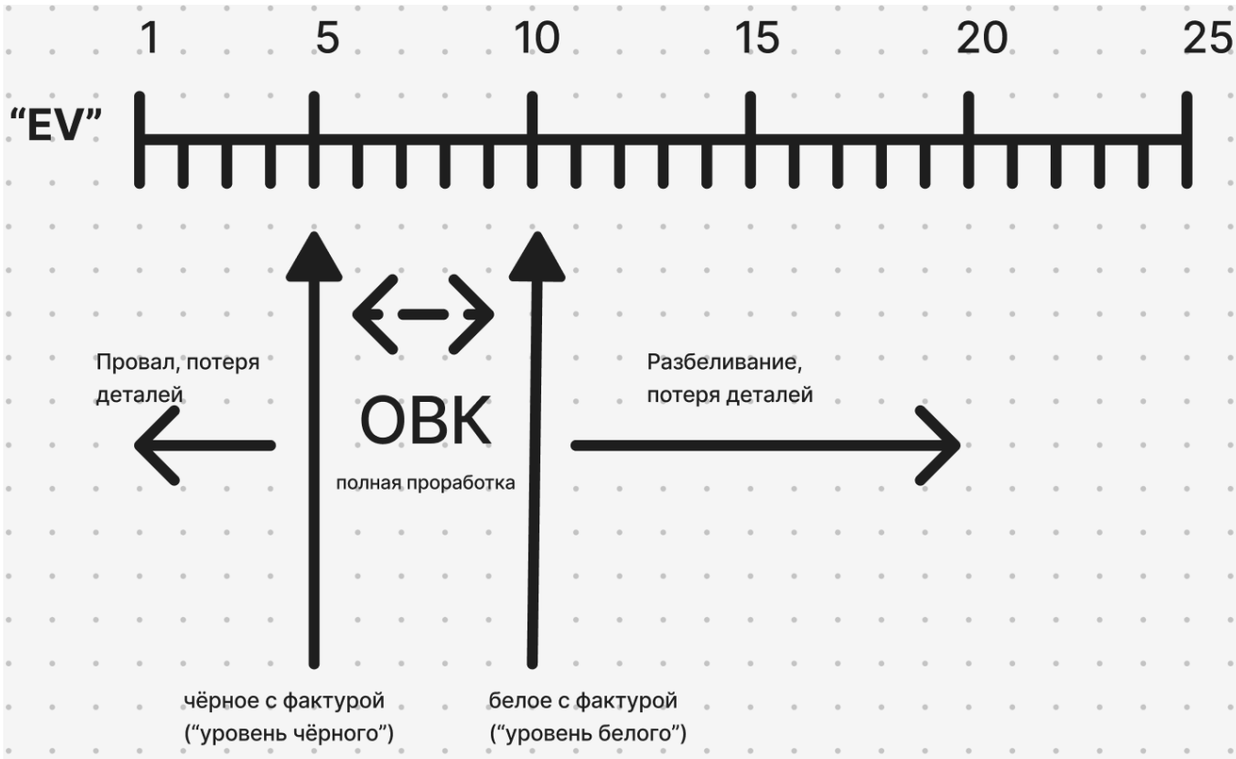
[7] The grisaille technique is a type of painting in which an image is created using various shades of the same color, most often gray or brown, and sometimes other neutral colors.

[8] A precision lighting system is a technique in which the luminous flux is divided into separate, precisely directed beams, each of which illuminates a strictly defined area of an object. This system uses specialized devices with blinds or tubes to precisely control the direction of light and allows you to create lighting with maximum detail and without unwanted shadows or highlights. The term "precision" means high precision in operation.

[9] The Fresnel lens on a cinema lighting device forms soft, directional lighting to create a natural light pattern on the stage. It is used for flood, contour or overhead lighting, creating a light spot with soft edges, which makes it easier to mix light from different sources. This is especially useful for lighting large areas of scenery or for spotlighting objects. The lens consists of a set of concentric grooves etched on a flat surface. This structure makes the lens much thinner and lighter than a standard convex lens of the same refractive power. The light passing through such grooves is focused into a narrow, directional beam, which then diffuses more gently than that of a standard spotlight.

Illustrations:

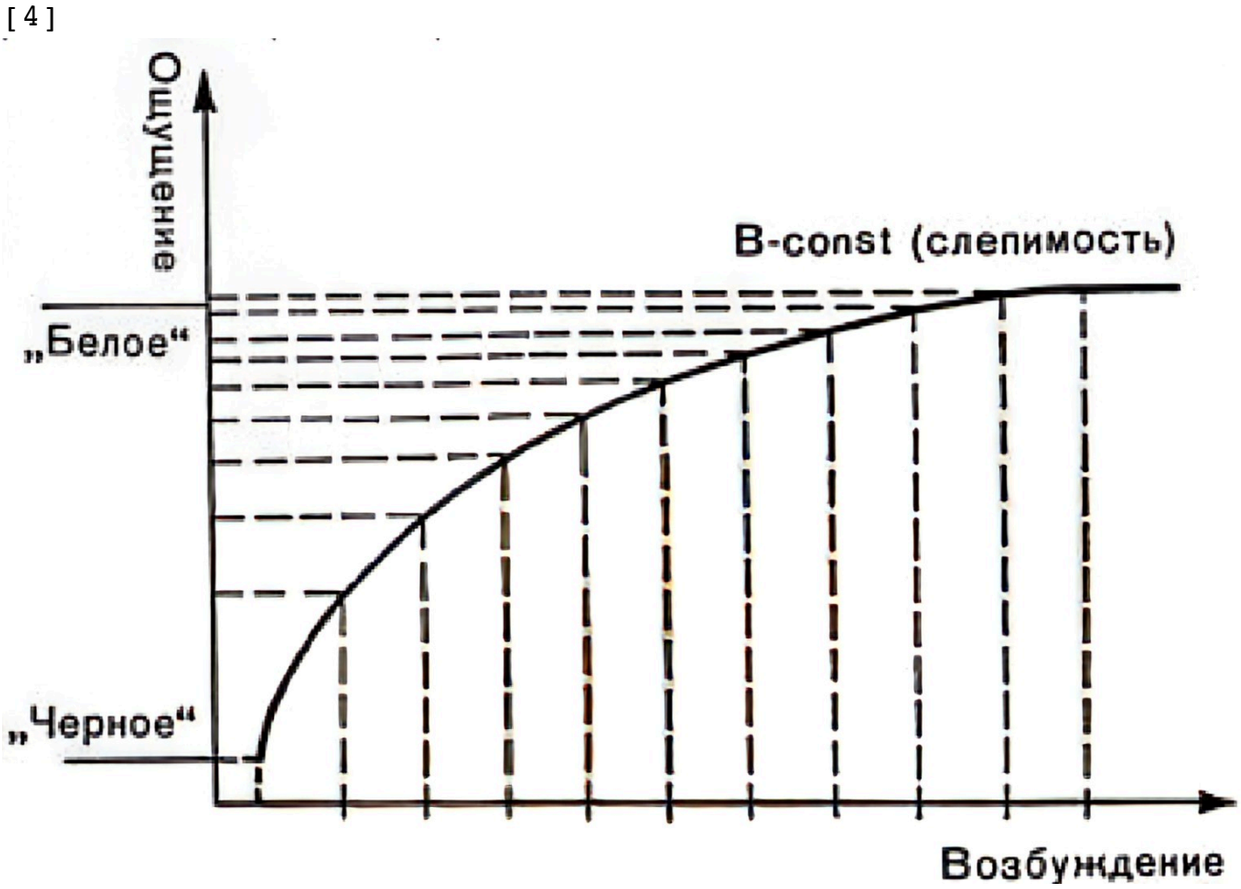
[1] Optimal visual contrast (OVC), EV - exposure units/relative brightness units


1
5
10
15
20
25
“EV”
Провал, потеря деталей
ОВК
полная проработка
Разбеливание, потеря деталей
чёрное с фактурой (“уровень чёрного”)
белое с фактурой (“уровень белого”)

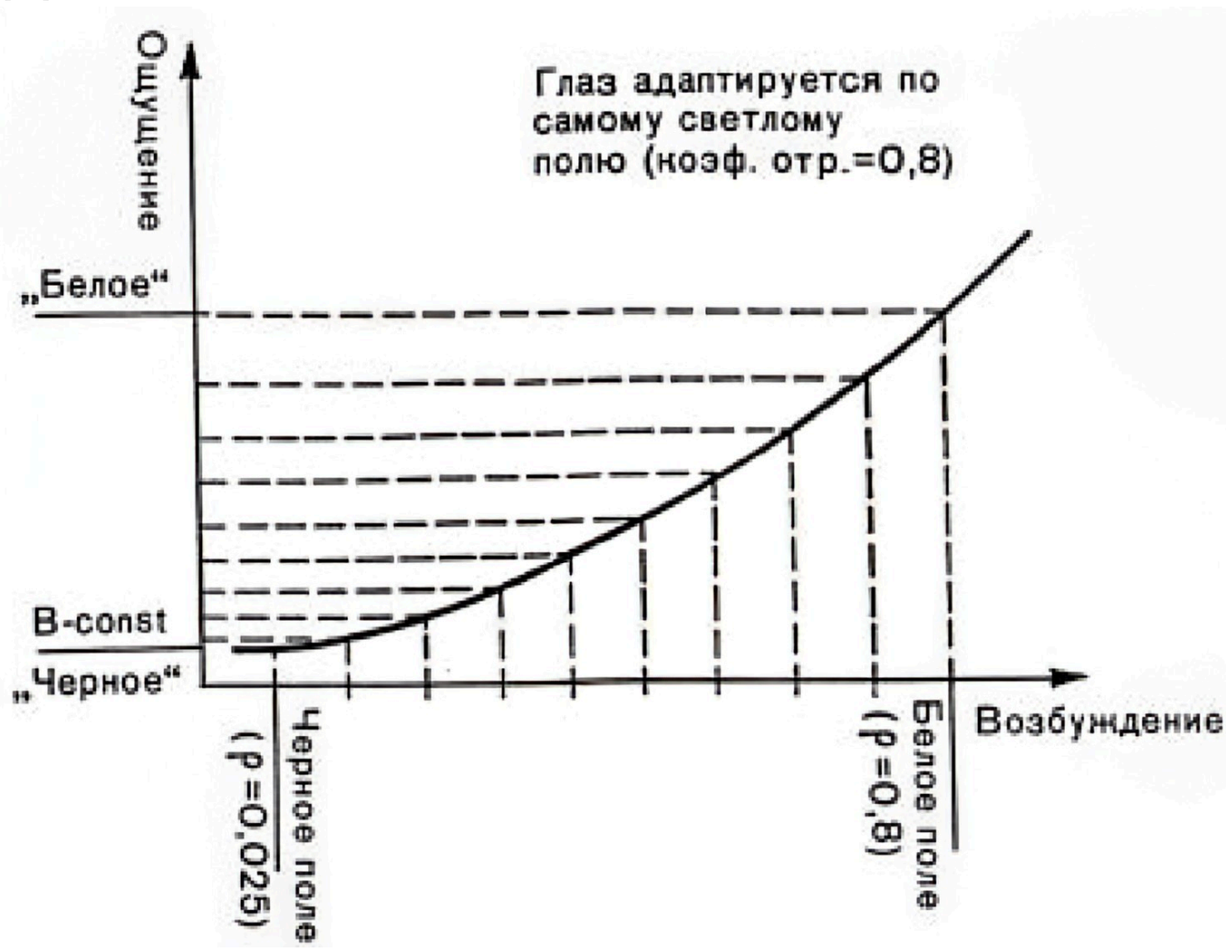
[2] Engraving "Hieronymus" by Albrecht Durer

[3] "Leningrad. - Rostral column at the Stock Exchange", A. P. Ostroumova

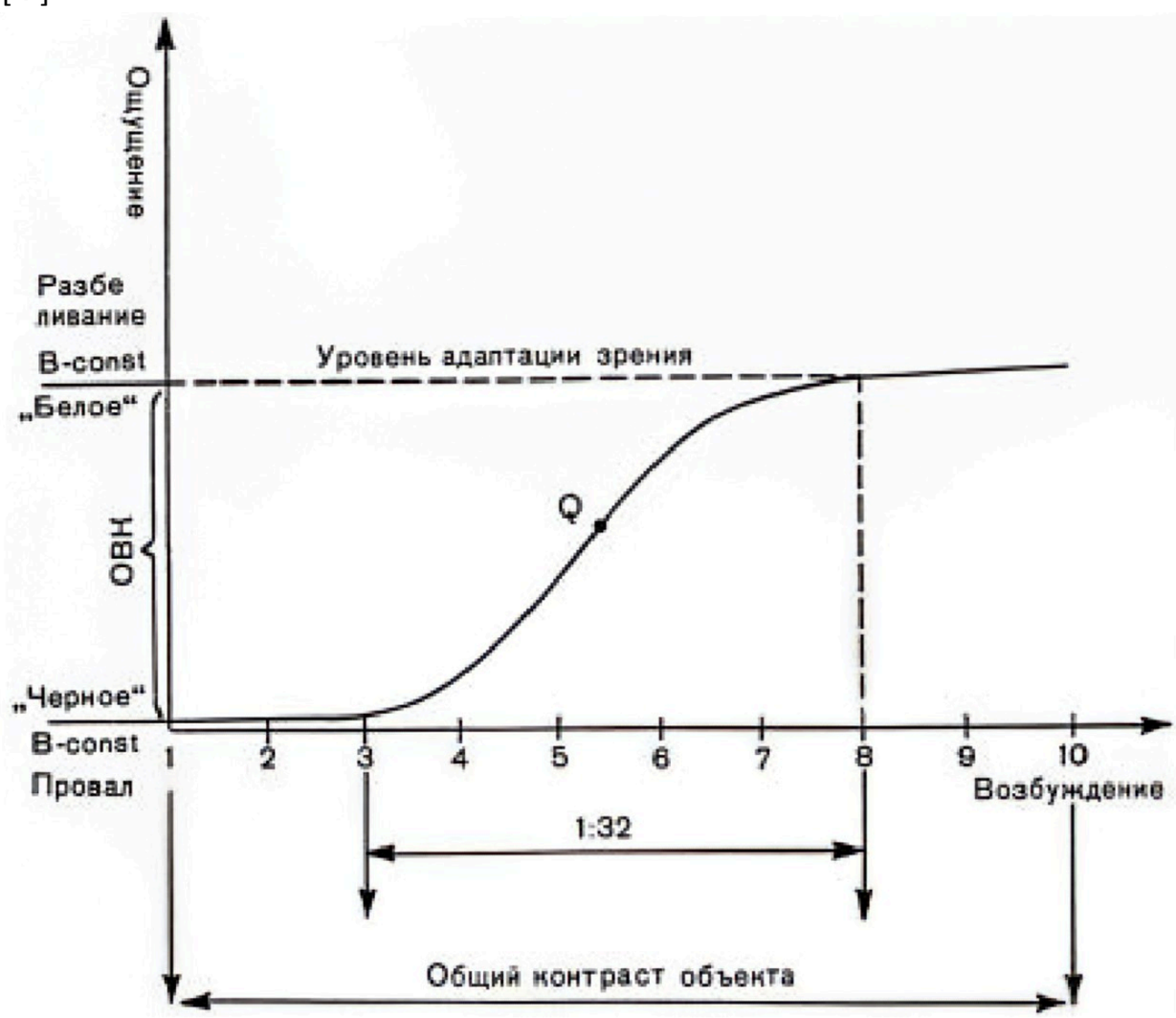
[4]



[5]

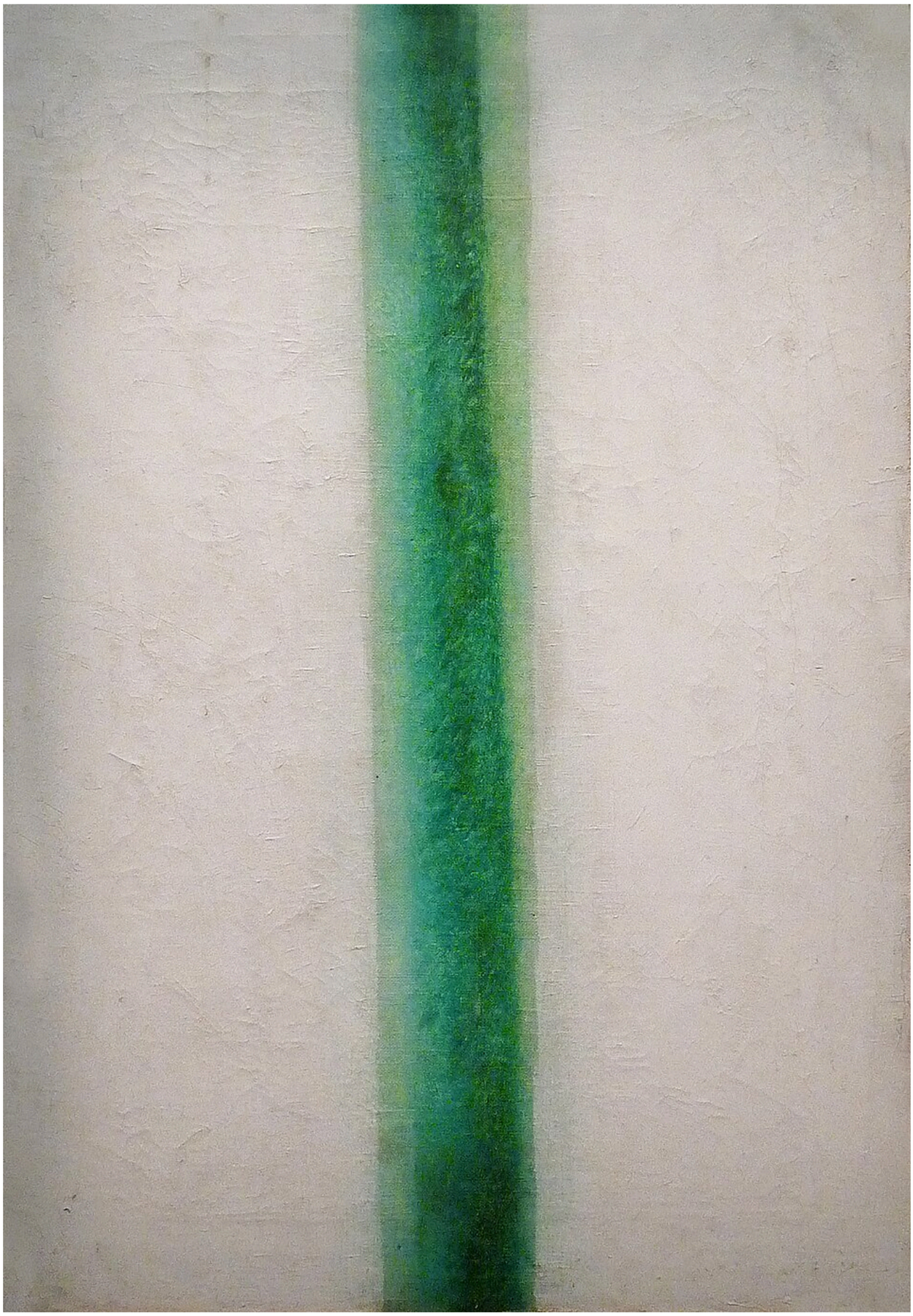

Ощущение
Глаз адаптируется по самому светлому полю (коэф. отр.=0,8)
„Белое“
B-const
„Черное“
Черное поле (ρ=0,025)
Белое поле (ρ=0,8)
Возбуждение

[6]


Ощущение
Разбе ливание
B-const
Уровень адаптации зрения
„Белое“
Q
ОВН
„Черное“
B-const
Провал
1
2
3
4
5
6
7
8
9
10
Возбуждение
1:32
Общий контраст объекта

[7] Simultaneous: "Three female figures", Kazimir Malevich

[8] Sequential: "The Green Stripe", Rozanova Olga

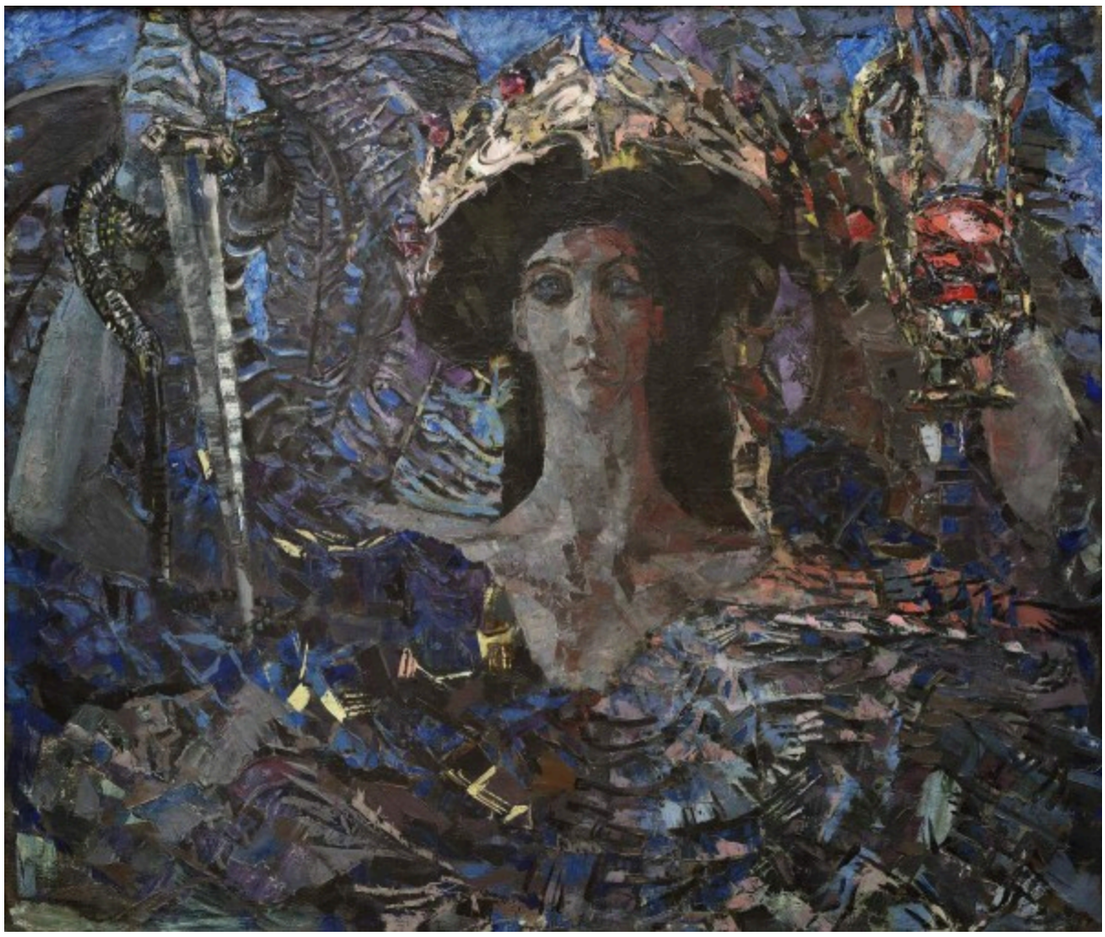
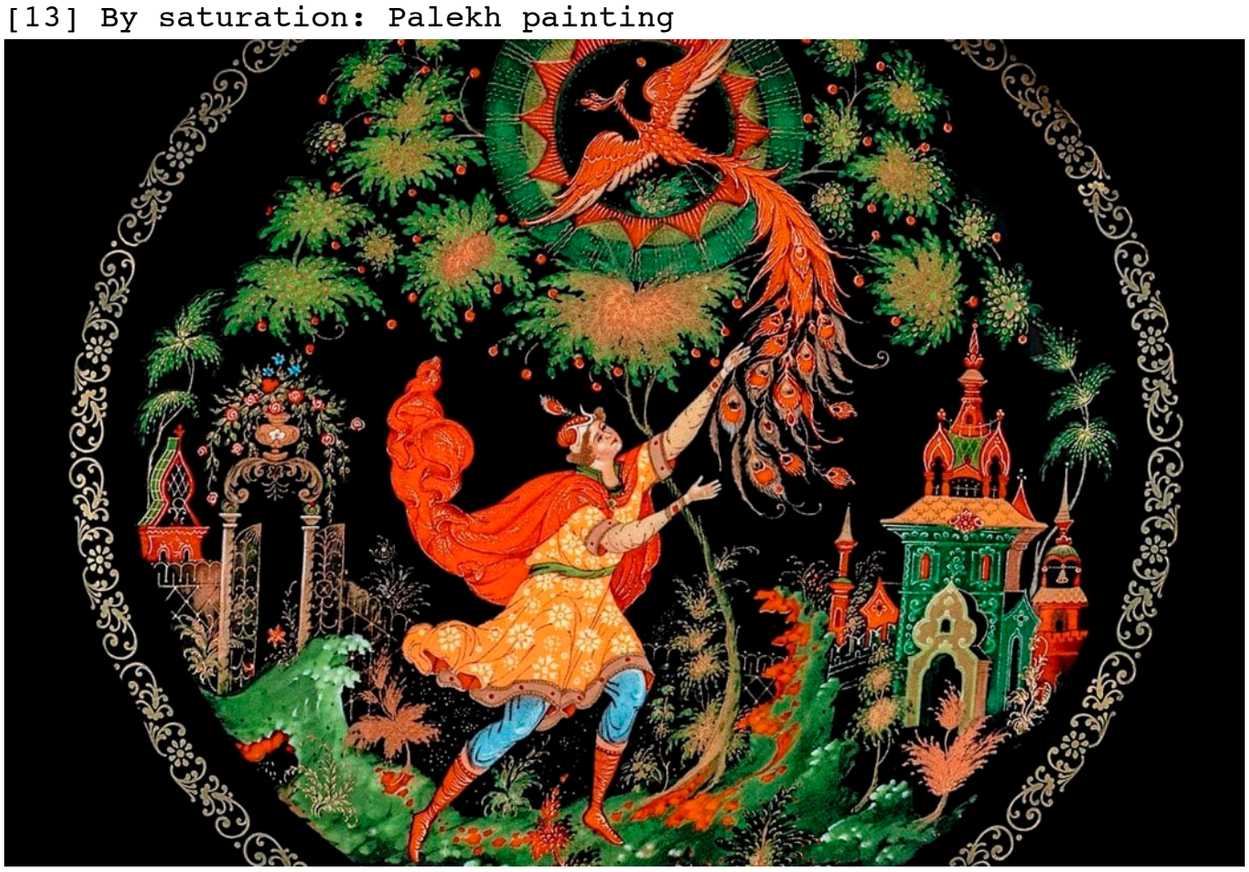


[9] Complementary: "Night Cafe" by Vincent Van Gogh

[10] Tint: "Still life with poppies and cornflowers", Ilya Mashkov

[11] Qualitative: "Saint Sebastian", Titian

[12] Quantitative: "The Six-Winged Seraphim", Mikhail Vrubel



[13] By saturation: Palekh painting